Shivanya Rathaur

BSc (Hons.) Data Science and AI, IIT Guwahati

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Github | LinkedIn | Portfolio

Summary

Final-year Data Science & AI student at IIT Guwahati, experienced across the analytics pipeline – SQL, Python-based EDA, and Power BI/Streamlit dashboarding. Built three end-to-end projects spanning 200K+ records, turning raw data into business-ready insights. Skilled in Python, SQL, MySQL, RAG, and prompt engineering.

Education

•Indian Institute of Technology, Guwahati *2023 – Present*
BSc (Hons.) Data Science and AI ; CGPA: 7.16 Guwahati, Assam

Projects

•Hotel Booking Demand Analysis
SQL, Python (Pandas, Matplotlib), Statistics, Power BI Github

- Analyzed ∼119,390 hotel reservations end-to-end using SQL for business queries and Python (Pandas, Matplotlib) for data cleaning, feature engineering, and exploratory data analysis
- Applied statistical tests to validate relationships (e.g., lead time vs. cancellation), uncovering a 27.53% overall cancellation rate driven by the Online TA channel (35.39% cancellation rate)
- Designed a 4-page interactive Power BI dashboard tracking key business KPIs – €34.45M estimated revenue, 87K bookings, €106.52 average ADR, and 79.98-days average lead time – across executive, cancellation, revenue, and seasonal views
- Delivered an executive report translating analytical findings into actionable recommendations for reducing cancellation-driven revenue loss and optimizing seasonal pricing

•Telecom Churn Intelligence: LLM-Powered BI Assistance
Python, Pandas, NumPy, Plotly, Streamlit, Google Gemini API, RAG Github

- Built an end-to-end BI assistant on the IBM Telco Customer Churn dataset (7,043 customers, 21 features), engineering a KPI analytics layer for churn (26.54%), retention, and revenue metrics
- Designed an interactive Streamlit dashboard with dynamic churn, contract, and service-level visualizations using Plotly for non-technical business stakeholders
- Implemented Retrieval-Augmented Generation (RAG) with the Google Gemini API and prompt engineering to let users query business metrics in natural language and auto-generate executive summaries, SWOT, and root-cause reports
- Identified key churn drivers – month-to-month contracts, absence of tech support/online security, and electronic-check payments – translating findings into retention recommendations

•India Historical Weather Analysis Dashboard
Python, Pandas, NumPy, Plotly, Streamlit Github

- Built an ETL pipeline and multi-page Streamlit dashboard analyzing 91,320 daily weather records across 10 major Indian cities over 25 years (2000–2024), automating deduplication, date standardization, and seasonal mapping
- Engineered derived features (temperature range, heatwave/heavy-rain flags) and built interactive Plotly visualizations – stacked histograms, box plots, radar charts – to surface climate outliers and regional variance
- Developed reactive time-series analysis with user-configurable date ranges and moving averages, plus an automated CSV export engine for filtered data slices
- Identified key climate patterns – 46.4°C historical peak in Delhi/Ahmedabad, 339.9mm single-day rainfall extreme in Ahmedabad, and wider seasonal variance in northern vs. southern cities

Technical Skills

- **Languages & Databases**: Python, SQL, MySQL
- **Libraries & Tools**: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Power BI, Streamlit, Excel, Git, VS Code, Jupyter Notebook
- **AI/ML**: Prompt Engineering, RAG, Google Gemini API

Certifications

- **Certifications**: Data, Data, Everywhere - Google (Coursera); Time Series Analysis - IIT Guwahati (Unstop)

Key courses taken

- **Mathematics & Statistics**: Linear Algebra, Introduction to Statistics, Statistical Inferencing
- **Data Science & Analytics**: Data Analysis Basics, Data Science: An Introduction, Data Mining & Warehousing, Data Modeling & Visualization, Time Series Analysis & Forecasting, Machine Learning Fundamentals
- **Programming & Databases**: Python Programming, R, Relational Database Management Systems